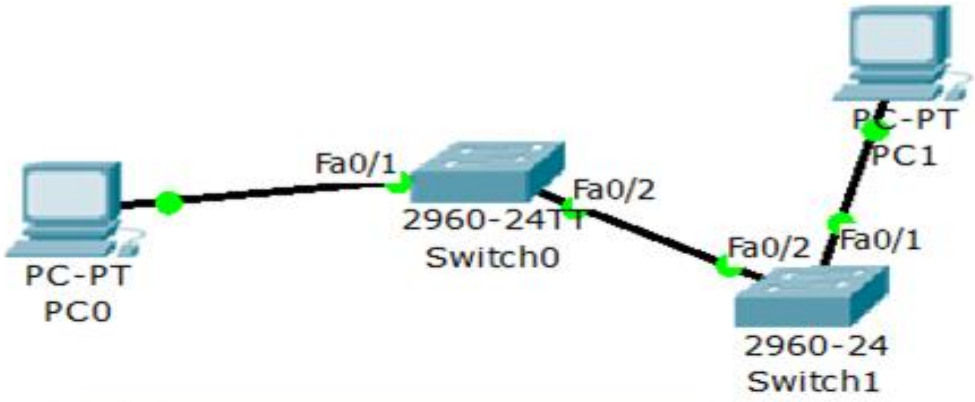
COU4305 Computer Networks and Security - Practical Session 1

PT Practical 1



Switch 0
Name : DMCSmainLab
IP of vlan 1 : 192.168.0.5 / 24
Management vlan 1

enable password : test
enable secret : mytest
console password : justwait

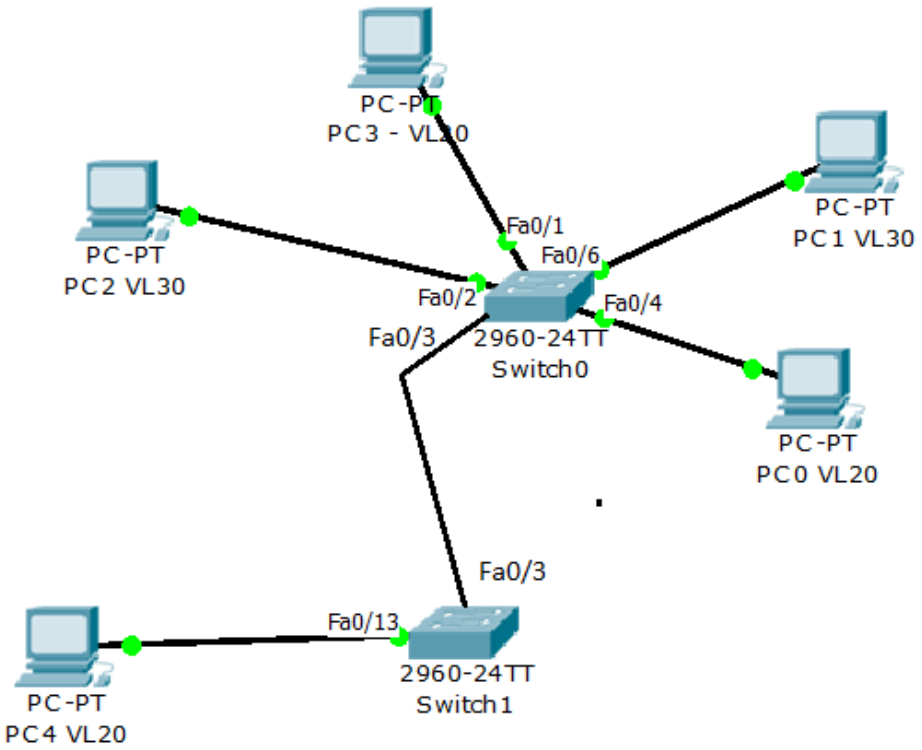
Login banner : Hi This is for authorised users

Switch 1
Name DmcsSpecialLab
IP of vlan 1: 192.168.0.10
Management Vlan 1

enable password : nishachara
enable secret : madhapawana
consolepassword :insecure
vty password : sirimayura

1. Assign a correct IPs to PC0 (192.168.0.1/24) and PC1 (192.168.0.2/24) then initiate ping operation.
2. Save Switch configurations

PT practical 2



Draw above topology in packet tracer. VL20 = 192.168.20.0/24, VL30 = 192.168.30.0/24 .

Assign the following ips to the PCs of the topology

VL20

PC0 IP 192.168.20.1

PC3 IP 192.168.20.2

PC4 IP 192.168.20.3

VL30

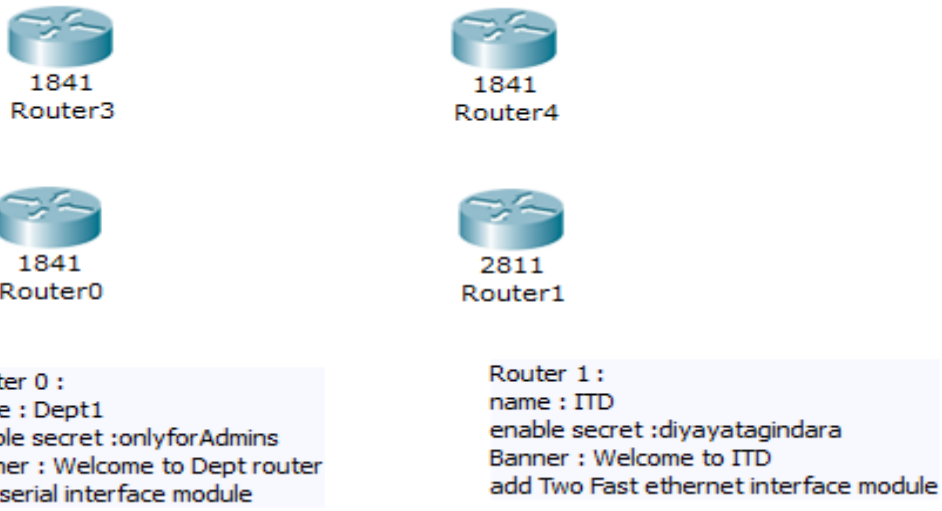
PC1 IP 192.168.30.1

PC2 IP 192.168.30.2

Configure VLANs, ping between PCs of same VLAN has to be successful, different VLAN unsuccessful.

Connect the Switch 0 and Switch 1 in such a way that we should be able to ping PC0 from PC4.

PT practical 3

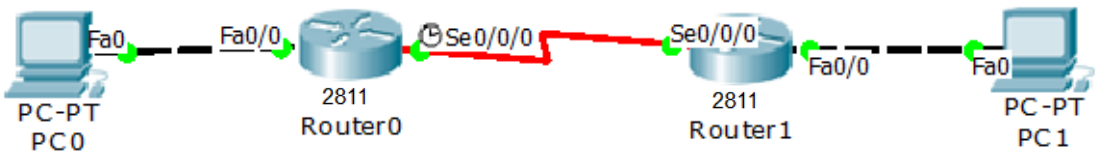


This is a basic router practical for configuring commands and adding interfaces modules. Do the tasks for router 0 and 1 and connect all 1841 routers to 2811 router using appropriate FE Wan connections

Router 0 : Use an interface card with one serial interface when adding the required interface

Router 1 - Use a card with two Fast Ethernet Wan Interfaces

PT practical 4



PC0 ip 192.168.0.2/24 and router interface of PC0 side 192.168.0.1/24

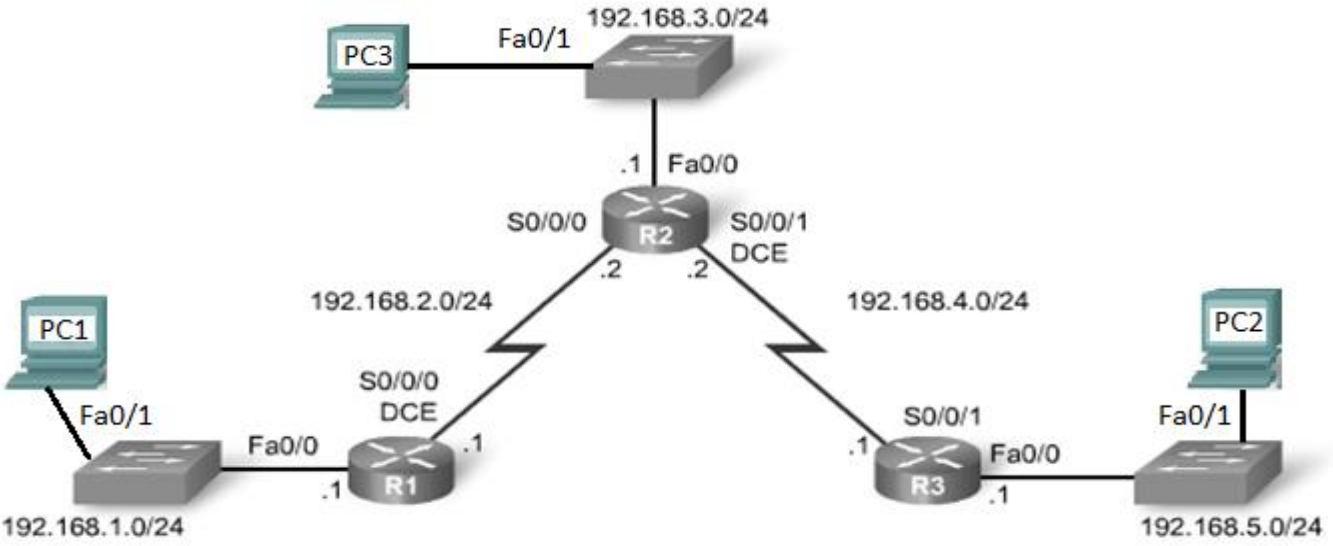
PC1 ip 192.168.2.2/24 and router interface of PC1 side 192.168.2.1/24

Router 0 serial interface ip 192.168.1.1/24, Router 1 serial interface 192.168.1.2, clock rate 64000

Based on the above information Configure the router with static routing to have correct ping from PC0 to PC1

When adding Serial interface Both routers, use high speed two port serial interface card.

PT practical 5



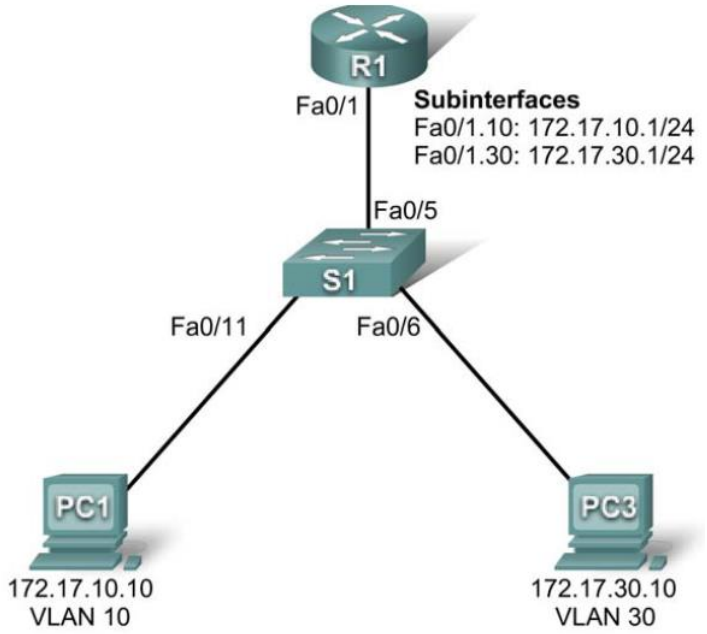
1. Configure the above topology and save a copy, then do the static routing
2. Configure dynamic routing algorithm (rip) in above topology

Use high speed serial interface card with 2 interfaces.

Use Fa0/2 of switch to connect the router fa0/0

.5 is the last octet value of PC IP address and clock rate for serial links are 64000.

PT practical 6 – Inter VLAN routing using router in Stick mode



1. Go to command line configuration of the switch and type **enable** to go to privileged mode

2. Go to configuration mode

S1#**configure terminal**

3. Create VLANs in switch

S1(config)#**vlan 10**
S1(config-vlan)#**vlan 30**

4. See the details of VLANs in switch
S1#**show vlan brief**

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	**Fa0/1.10**	172.17.10.1	255.255.255.0	N/A
	Fa0/1.30	172.17.30.1	255.255.255.0	N/A
PC1	**NIC**	172.17.10.10	255.255.255.0	172.17.10.1
PC3	**NIC**	172.17.30.10	255.255.255.0	172.17.30.1

5. Then we have to assign ports to VLAN. We assign Fa0/11 port first

S1(config-if)#**interface fa0/11**
S1(config-if)#**switchport mode access**
S1(config-if)#**switchport access vlan 10**

6. Repeat the steps for the Fa0/6 interface for VLAN 30.

S1(config)#**interface fa0/6**
S1(config-if)#**switchport mode access**
S1(config-if)#**switchport access vlan 30**

7. Make the port connected to router as trunk port because we are sending the multiple VLAN traffic on the same port to the router
S1(config-if)#**interface fa0/5**
S1(config-if)#**switchport mode trunk**
S1(config-if)#**switchport trunk native vlan 1**
S1(config-if)#**switchport trunk allowed vlan add 10**
S1(config-if)#**switchport trunk allowed vlan add 30**
S1(config-if)#**end**

8.Then we have to configure the router to have virtual subinterfaces accepting the traffic coming in different VLANs

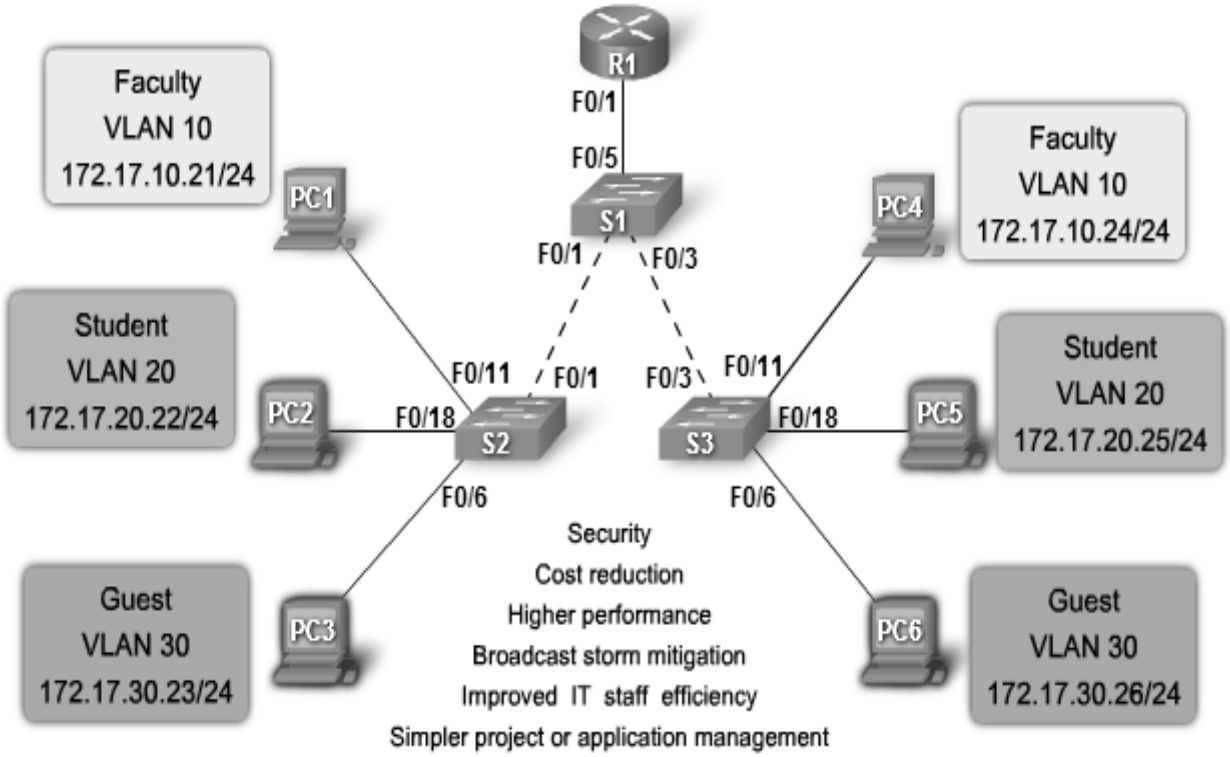
R1(config)#**interface fa0/1.10**

R1(config-subif)**#encapsulation dot1Q 10**
R1(config-subif)**#ip address 172.17.10.1 255.255.255.0**
R1(config-subif)**#no shutdown**
R1(config-subif)**#interface fa0/1.30**
R1(config-subif)**#encapsulation dot1Q 30**
R1(config-subif)**#ip address 172.17.30.1 255.255.255.0**
R1(config-subif)**#no shutdown**

9. Finally turn on main interface of the router that has sub interfaces.
R1(config)**#interface fa0/1**
R1(config-if)**#no shutdown**
R1(config-if)**#end**

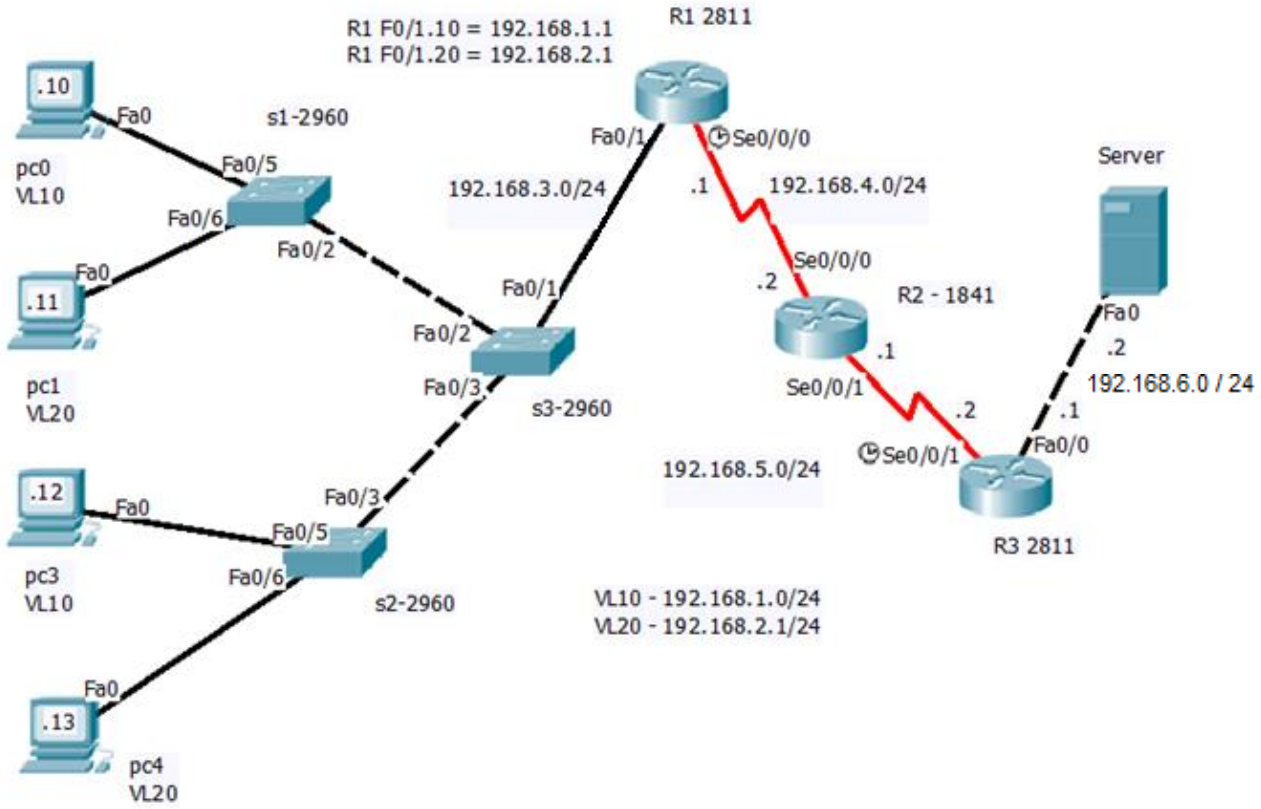
10. Ping from one PC to another PC in different VLAN it should be successful.

PT practical 7



Configure the above setup without router; at the end of this class you should be able to configure it with a router in stick mode to transfer inter-vlan traffic.

PT practical 8



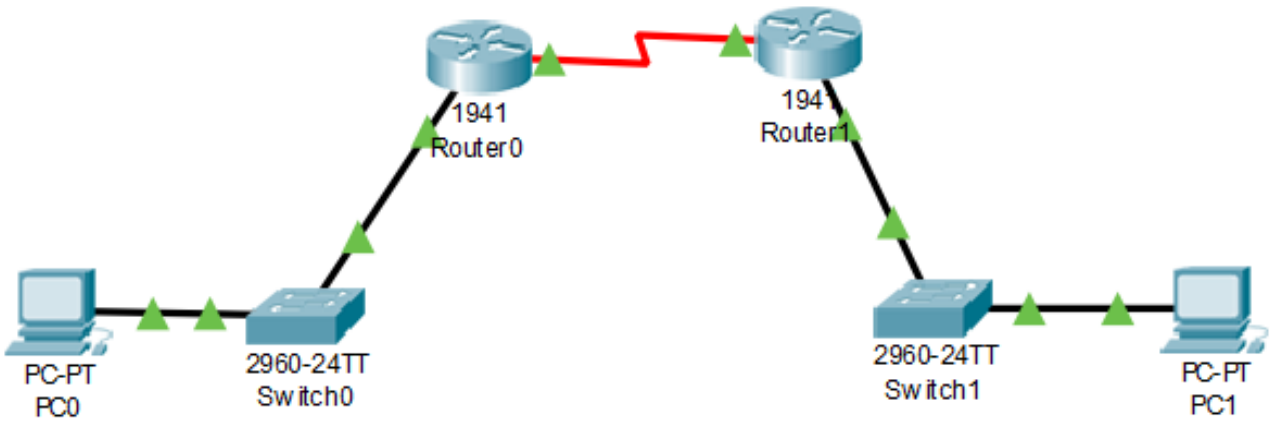
Always use highspeed 2 port seria interfacel card for all the routers and use Static Routing

PT practical 9

Dynamic routing protocol EIGRP (CISCO proprietary) configuration.

It can be done in three ways

1. Using classful lip address
2. Using subnet mask
3. Using wildcard mask



- **Classfull ip addresses**

  Configuration commands

First go to the cli of **Router0** and do the following configurations

```
Router> enable
Router# configure terminal
```

Enter the **serial port** and do the following configurations

```
Router(config)#interface serial 0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#bandwidth 64
Router(config-if)#no shutdown
Router(config-if)#exit
```

Enter the **gigabyte ethernet port** and do the following configurations

```
Router(config)#interface gig0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
```

Then go to the cli of **Router1** and do the following configurations

```
Router> enable
Router# configure terminal
```

Enter the **serial port** and do the following configurations

```
Router(config)#interface serial 0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#bandwidth 64
Router(config-if)#no shutdown
Router(config-if)#exit
```

Enter the **gigabyte ethernet port** and do the following configurations

```
Router(config)#interface gig0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
```

Now add the networks which were connected to the routers using **classful ip addresses**

Router 0

```
Router> enable
Router# configure terminal
Router(config)# router eigrp 5
Router(config-router)# network 192.168.1.0
Router(config-router)# network 192.168.2.0
```

Router 1

```
Router> enable
Router# configure terminal
Router(config)# router eigrp 5
Router(config-router)# network 192.168.1.0
Router(config-router)# network 192.168.3.0
```

Now check the routing tables of both routers and understand the routing path detection of EIGRP.

- **Classless ip addresses**

1. **Using subnetmask of the networks**

Router 0

```
Router> enable
Router# configure terminal
Router(config)# router eigrp 5
Router(config-router)# network 172.168.1.0 255.255.255.0
Router(config-router)# network 172.168.2.0 255.255.255.0
```

Router 1

```
Router> enable
Router# configure terminal
Router(config)# router eigrp 5
Router(config-router)# network 172.168.1.0 255.255.255.0
Router(config-router)# network 172.168.3.0 255.255.255.0
```

Now check the routing tables of both routers and understand the routing path detection of EIGRP.

2. **Using wildcard mask of the networks**

Router 0

```
Router> enable
Router# configure terminal
Router(config)# router eigrp 5
Router(config-router)# network 172.168.1.0 0.0.0.255
Router(config-router)# network 172.168.2.0 0.0.0.255
```

Router 1

```
Router> enable
Router# configure terminal
Router(config)# router eigrp 5
Router(config-router)# network 172.168.1.0 0.0.0.255
Router(config-router)# network 172.168.3.0 0.0.0.255
```